MATHS MANTHAN
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CBSE CLASS 10 | MATHEMATICS

STATISTICS

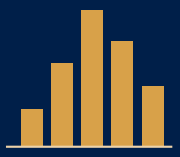
Master Concept, Formula & Visual Problem-Solving Handbook

Academic Year 2026–27

Organise the data. Master the intervals.
Calculate with confidence.

Direct, Assumed & Step-Deviation Mean | Mode | Median | Missing Frequencies

A student-friendly Maths Manthan revision resource



What this handbook includes

- Complete derivations & formulas for Mean ($\bar{x}$), Mode, and Median of grouped data.
- All three Mean techniques: Direct, Assumed Mean (d_i), and Step-Deviation (u_i).
- An optional approximate empirical relationship connects Mean, Median, and Mode.
- Double missing frequency (x, y) master models and exam trap-busters.

How to use it

Check class intervals before applying grouped Median/Mode formulas. State the symbolic formula before substituting values. For Median, use the cumulative frequency from the preceding class; a grouped Median may lie on a class boundary. For a unique modal class, check that the estimated Mode lies within that class.

Official syllabus scope (CBSE 2026–27)

The published CBSE Class X Mathematics curriculum for 2026–27 lists **Mean, Median, and Mode of grouped data**; it specifies direct, assumed-mean, and step-deviation methods for Mean and says bimodal situations are to be avoided. This handbook also uses class-boundary conversion and cumulative-frequency tables as supporting tools for calculations; their inclusion is not a claim that each is a separate prescribed syllabus item.

Source: CBSE Mathematics, Class X (2026–27), Unit VII: Statistics and Probability.

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Prepared for learning and revision – not an official CBSE publication.

CHAPTER 13

SUPPORTING SKILL

STATISTICAL FOUNDATIONS

Data Types, Continuous Boundaries & Class Interval Mechanics

Module 1 | Progression of Data: Raw to Grouped Frequency

Statistics deals with the collection, presentation, analysis, and interpretation of numerical data.

1. Raw Data

Individual observations:

18, 25, 12, 25, 30, 18, 25, 42

2. Frequency Count

Count each distinct value:

$12 \to 1,\ 18 \to 2,\ 25 \to 3,\ 30 \to 1,\ 42 \to 1$

3. Grouped Data

Half-open class intervals:

$[10, 20)$	3
$[20, 30)$	3
$[30, 40)$	1
$[40, 50)$	1

Module 2 | Anatomy of a Class Interval

For a class interval with lower limit L and upper limit U:

- **Lower Class Limit (L):** The lowest data value designated for that class.
- **Upper Class Limit / Boundary (U):** The upper edge of the interval. In a half-open class $[L, U)$, U is excluded from the class and serves as its upper boundary.
- **Class Width / Class Size (h):** The difference between the true upper and lower class boundaries:

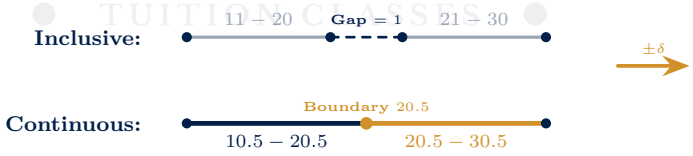
$$\mathbf{h} = \text{Upper Boundary} - \text{Lower Boundary}$$

- **Class Mark (Midpoint x_i):** The central representative value of the class interval:

$$\mathbf{x_i} = \frac{\text{Lower Limit} + \text{Upper Limit}}{\mathbf{2}}$$

Module 3 | Discontinuous (Inclusive) vs. Continuous (Exclusive) Classes

WHEN CONTINUOUS CLASS BOUNDARIES ARE NEEDED:
The grouped Median and standard grouped Mode formulas used here require continuous class intervals. Convert inclusive integer classes such as 11–20, 21–30 to continuous boundaries before substituting. The standard grouped Mode formula also assumes equal class widths; where widths differ, use the appropriate frequency-density/histogram approach if prescribed.



The Universal Continuity Conversion Rule:

1. Find the uniform gap g = lower limit of next class − upper limit of current class.
2. Use the correction $\delta = \frac{g}{2}$. For 11–20 and 21–30, $g = 1$ and $\delta = 0.5$.
3. Set $L_{\text{new}} = L - \delta$ and $U_{\text{new}} = U + \delta$.

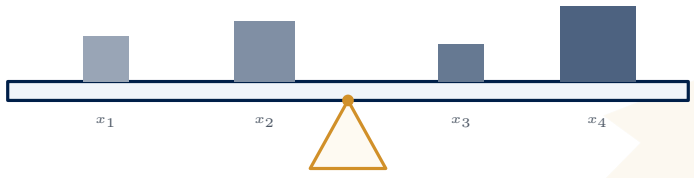
In this example, 11–20 *becomes* 10.5–20.5. *The class mark remains* 15.5 *and the width is* 10.

CENTRAL TENDENCY I | The Arithmetic Mean: Concept & Direct Method

CORE SYLLABUS

Module 4 | The Physics of Mean: Center of Gravity / Fulcrum

The arithmetic mean ($\bar{x}$) represents the **central balance point** of a distribution. If each observation x_i represents a position on a massless beam and each frequency f_i represents a physical weight placed at that point, the fulcrum must be placed exactly at $\bar{x}$ to keep the system in equilibrium:



$$\sum \mathbf{f_i(x_i - \bar{x}) = 0} \implies \textbf{Total Left Torque} = \textbf{Total Right Torque}$$

Module 5 | Method 1: The Direct Method

The direct method computes the mean by taking the weighted sum of all class marks and dividing by the total frequency:

Direct Method Mean Equation:

$$\bar{\mathbf{x}} = \frac{\sum_{\mathbf{i=1}}^{\mathbf{n}} \mathbf{f_i x_i}}{\sum_{\mathbf{i=1}}^{\mathbf{n}} \mathbf{f_i}} = \frac{\sum \mathbf{f_i x_i}}{\mathbf{N}}$$

Where:

- $x_i = \frac{\text{Lower Limit}+\text{Upper Limit}}{2}$ is the class mark of the i-th class interval.
- f_i is the frequency of the i-th class interval.
- $N = \sum f_i$ is the total number of observations.

Module 6 | Standard 4-Column Table Format (Direct Method)

Class Interval	Frequency (f_i)	Class Mark (x_i)	Product ($f_i x_i$)
$0 - 10$	f_1	5	$f_1 \times 5$
$10 - 20$	f_2	15	$f_2 \times 15$
$20 - 30$	f_3	25	$f_3 \times 25$
$30 - 40$	f_4	35	$f_4 \times 35$
Total	$\sum \mathbf{f_i = N}$	—	$\sum \mathbf{f_i x_i}$

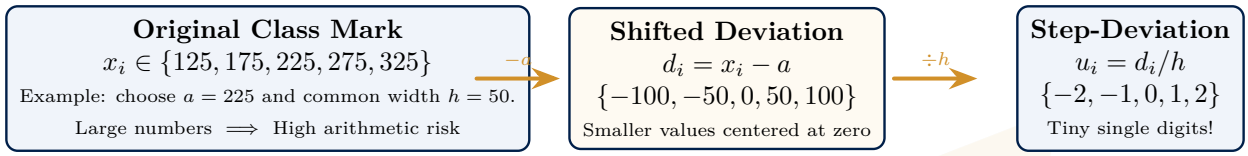
When to use the Direct Method:
The direct method is convenient when the products $f_i x_i$ are easy to calculate. If class marks are large or products become cumbersome, the assumed-mean method can reduce arithmetic. The step-deviation method is especially convenient when all class intervals have a common width.

CENTRAL TENDENCY I | Mean via Assumed Mean & Step-Deviation

CORE SYLLABUS

Module 7 | The Mathematics of Shift & Scale: Why Deviations Work

When class marks x_i are large, multiplying $f_i x_i$ is tedious. We simplify calculations using two linear transformations:



Module 8 | Method 2: The Assumed Mean Method (Deviation Method)

Choose a central class mark as the **Assumed Mean** (a). Compute deviations $d_i = x_i - a$:

Assumed Mean Formula:

$$\bar{\mathbf{x}} = \mathbf{a} + \frac{\sum \mathbf{f_i d_i}}{\sum \mathbf{f_i}} = \mathbf{a} + \bar{\mathbf{d}}$$

Where:

- a is the assumed mean (chosen from the middle of the x_i column).
- $d_i = x_i - a$ is the deviation of each class mark from a.
- $\bar{d} = \dfrac{\sum f_i d_i}{\sum f_i}$ is the average deviation.

Module 9 | Method 3: The Step-Deviation Method

When all class intervals have equal width h, all deviations d_i share a common factor h. We scale down by h:

Step-Deviation Formula:

$$\bar{\mathbf{x}} = \mathbf{a} + \left(\frac{\sum \mathbf{f_i u_i}}{\sum \mathbf{f_i}} \right) \times \mathbf{h}$$

Where:

- $u_i = \frac{x_i - a}{h} = \frac{d_i}{h}$.
- h is the common class width.
- **Gold Rule:** Never forget to multiply the average step-deviation by h before adding a!

Module 10 | Fast Decision Matrix: Which Method to Choose in Exams?

Method	When to Choose	Key Advantage / Caution
1. Direct Method	When products $f_i x_i$ are manageable.	Simple formula; products may be larger.
2. Assumed Mean	When class marks are large or products are cumbersome; also works with unequal widths.	Uses deviations from a to reduce arithmetic.
3. Step-Deviation	When all class intervals have equal width h and step deviations are convenient.	Scales deviations down; often efficient for hand calculation.

WORKED MODEL I | Step-Deviation Mean & Missing Frequencies

WORKED MODEL

Model 1 | Complete Step-Deviation Mean Solution

Problem: Calculate the mean daily pocket allowance of children in a locality from the following data:

Pocket Allowance (Rs.)	f_i	x_i	$d_i = x_i - 18$	$u_i = d_i/2$	$f_i u_i$
$11-13$	7	12	-6	-3	$7 \times (-3) = -21$
$13-15$	6	14	-4	-2	$6 \times (-2) = -12$
$15-17$	9	16	-2	-1	$9 \times (-1) = -9$
$17-19$	13	**18 (a)**	0	0	$13 \times 0 = 0$
$19-21$	20	20	$+2$	$+1$	$20 \times 1 = 20$
$21-23$	5	22	$+4$	$+2$	$5 \times 2 = 10$
$23-25$	4	24	$+6$	$+3$	$4 \times 3 = 12$
Total	$\sum \mathbf{f_i} = \mathbf{64}$	—	—	—	$\sum \mathbf{f_i u_i} = \mathbf{0}$

Step-by-Step Calculation:

1. Assumed Mean $a = 18$, Class size $h = 2$.
2. Sum of positive products: $20 + 10 + 12 = +42$.
3. Sum of negative products: $-21 - 12 - 9 = -42 \implies \sum \mathbf{f_i u_i} = +\mathbf{42} - \mathbf{42} = \mathbf{0}$.
4. Applying the Step-Deviation Formula:

$$\bar{\mathbf{x}} = \mathbf{a} + \left(\frac{\sum \mathbf{f_i u_i}}{\sum \mathbf{f_i}}\right) \times \mathbf{h} = \mathbf{18} + \left(\frac{\mathbf{0}}{\mathbf{64}}\right) \times \mathbf{2} = \mathbf{18}$$

Answer: The mean daily pocket allowance is **Rs. 18.00**.

Model 2 | Protocol for Missing Frequency Problems

When the mean ($\bar{x}$) is already given in the question and one or two frequencies are missing (f or f_1, f_2):

The 2-Equation Setup for Double Missing Frequencies (f_1, f_2):

- **Equation 1 (From Total Frequency N):**

$$\sum f_i = \text{Sum of known frequencies} + f_1 + f_2 = N \implies \mathbf{f_1 + f_2 = N - \sum f_{known}}$$

- **Equation 2 (From Mean $\bar{x}$):**

$$\bar{x} = \frac{\sum f_i x_i}{N} \implies \sum \mathbf{f_i x_i = N \cdot \bar{x}}$$

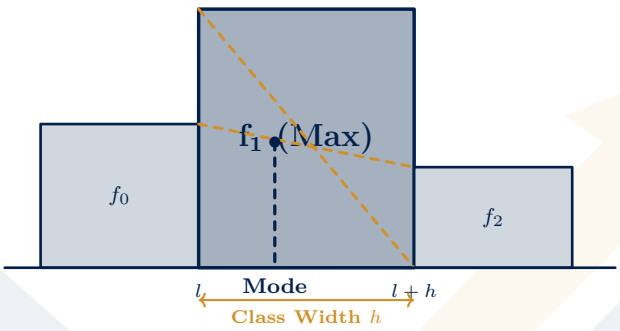
  Express $\sum f_i x_i$ algebraically in terms of f_1 and f_2, then solve simultaneously!

CENTRAL TENDENCY II | Mode: The Peak Frequency Measure

CORE SYLLABUS

Module 11 | Geometric Meaning: The Histogram Peak

The **Mode** is the observation having the maximum frequency. In grouped data, we first locate the **Modal Class** (the interval with the highest frequency), and then interpolate within that interval using the relative pull of the preceding (f_0) and succeeding (f_2) frequencies:



Module 12 | Mode Formula Suite for Grouped Data

Mode Formula:

$$\text{Mode} = \mathbf{l} + \left(\frac{\mathbf{f_1} - \mathbf{f_0}}{\mathbf{2f_1} - \mathbf{f_0} - \mathbf{f_2}} \right) \times \mathbf{h}$$

Anatomy of Variables (Remember the sequence: $0 \to 1 \to 2$):

- l = Lower limit of the modal class.
- h = Class size (width) of the modal class.
- f_1 = Frequency of the modal class (the highest frequency in the table).
- f_0 = Frequency of the class **preceding** the modal class (one row above).
- f_2 = Frequency of the class **succeeding** the modal class (one row below).

Module 13 | Step-by-Step Mode Protocol & Validation Check

1. **Check Continuity:** Ensure the class intervals are continuous ($10-20, 20-30, \ldots$).
2. **Identify f_1:** Scan the frequency column and locate the maximum frequency value.
3. **Mark Modal Class:** The class interval corresponding to f_1 is the modal class. Extract l and h.
4. **Read Neighbors:** Read f_0 directly above f_1, and f_2 directly below f_1.
5. **Reality Check:**

EXAM STRATEGY

For a unique modal class and the standard grouped Mode model, the estimate should fall within the modal class. If it falls outside, recheck the modal class, neighboring frequencies, and arithmetic.

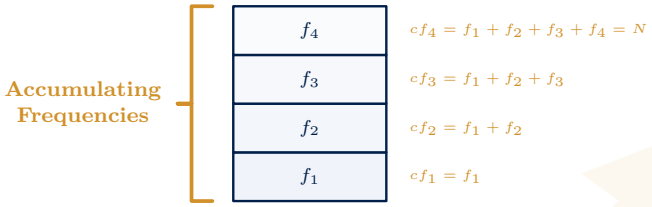
MATHS MANTHAN TUITION CLASSES

CUMULATIVE FREQUENCY | "Less Than" and "More Than" Distributions

SUPPORTING SKILL

Module 14 | The Cumulative Frequency Running Ladder

Cumulative Frequency (cf) represents the running total of frequencies up to a specified boundary:



Module 15 | "Less Than" vs. "More Than" Cumulative Frequency Tables

Standard Distribution		"Less Than" Type (Upper Limits)		"More Than" Type (Lower Limits)	
Class	f_i	**Marks**	cf	**Marks**	cf
$0 - 10$	5	Less than 10	5	More than or equal to 0	50 (N)
$10 - 20$	8	Less than 20	$5 + 8 = 13$	More than or equal to 10	$50 - 5 = 45$
$20 - 30$	12	Less than 30	$13 + 12 = 25$	More than or equal to 20	$45 - 8 = 37$
$30 - 40$	15	Less than 40	$25 + 15 = 40$	More than or equal to 30	$37 - 12 = 25$
$40 - 50$	10	Less than 50	$40 + 10 = 50$	More than or equal to 40	$25 - 15 = 10$

Module 16 | Reverse Engineering: Converting Cumulative Tables to Intervals

A cumulative-frequency table can be converted back to ordinary class frequencies by subtraction. This is a supporting skill when cumulative totals are supplied and a grouped-data calculation is required:

Recovering Frequencies by Successive Subtraction:

- First class frequency: $f_1 = cf_1$.
- i-th class frequency: $\mathbf{f_i = cf_i - cf_{i-1}}$.
- **Sanity Check:** The sum of all recovered frequencies must equal the final cumulative frequency (N).

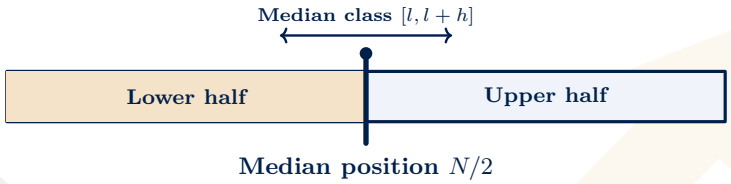
MATHS MANTHAN TUITION CLASSES

CENTRAL TENDENCY III | Median: The 50–50 Positional Measure

CORE SYLLABUS

Module 17 | Positional Partitioning: The $\frac{N}{2}$ Rule

For ordered raw data, the **median** is the middle value (or the average of the two middle values when the number of observations is even). For grouped data, the formula estimates the 50th percentile by interpolation. At least half the observations are at or below the median and at least half are at or above it; with ties, the proportions strictly below and strictly above need not each be exactly 50%.



Module 18 | Median Formula Suite for Grouped Data

Median Formula:

$$\text{Median} = \mathbf{l} + \left(\frac{\mathbf{\frac{N}{2} - cf}}{\mathbf{f}} \right) \times \mathbf{h}$$

Anatomy of Variables (Crucial Distinction):

- $N = \sum f_i$ is the total frequency.
- $\frac{N}{2}$ is half of the total frequency.
- **Median Class:** The first class whose cumulative frequency is **strictly greater than** $\frac{N}{2}$. If a cumulative frequency equals $N/2$ exactly at the preceding boundary, the grouped Median can lie on the shared class boundary.
- l = Lower limit of the median class.
- **cf** = Cumulative frequency of the class **preceding** the median class (one row above!).
- **f** = Frequency of the median class itself (from the frequency column).
- h = Class width of the median class.

THE #1 BLUNDER IN CLASS 10 STATISTICS:
cf DOES NOT COME FROM THE MEDIAN CLASS!
A common substitution error is to use the cumulative frequency of the median class itself.

TRAP ALERT

Remember: In $\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$:

- f comes from the **Median Class** (current row).
- cf comes from the **Preceding Class** (row above)!

WORKED MODEL II | Double Missing Frequency (x, y) Master Problem

WORKED MODEL

Model 3 | CBSE-Style Missing-Frequency Model ($N = 60$, Median $= 28.5$)

Problem: The median of the following frequency distribution is 28.5. Find the values of x and y if total frequency is 60:

Class Interval	Frequency (f_i)	Cumulative Frequency (cf)
$0 - 10$	5	5
$10 - 20$	x	$5 + x$
20 – 30 (Median Class)	**20** (**f**)	**25 + x**
$30 - 40$	15	$40 + x$
$40 - 50$	y	$40 + x + y$
$50 - 60$	5	$45 + x + y$
Total	**N = 60**	—

Step-by-Step Algebraic Solution:

1. **Equation 1 (From Total Frequency N):**

$$45 + x + y = 60 \implies x + y = 60 - 45 \implies \mathbf{x + y = 15} \quad \text{— (Eq. 1)}$$

2. **Locate Median Class from Given Median:** Given Median = 28.5. The value 28.5 lies inside the interval **20 – 30**.

Therefore: Median Class = **20 – 30**

3. **Extract Formula Parameters:**

$$l = 20, \quad \frac{N}{2} = \frac{60}{2} = 30, \quad f = 20, \quad h = 10$$

$$\mathbf{cf = 5 + x} \quad \text{(from the preceding row } 10 - 20!)$$

4. **Substitute into the Median Formula:**

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$$

$$28.5 = 20 + \left(\frac{30 - (5 + x)}{20} \right) \times 10$$

$$8.5 = \frac{25 - x}{2} \implies 2 \times 8.5 = 25 - x \implies 17 = 25 - x$$

$$\mathbf{x = 25 - 17 = 8}$$

5. **Find y using Equation 1:**

$$8 + y = 15 \implies \mathbf{y = 15 - 8 = 7}$$

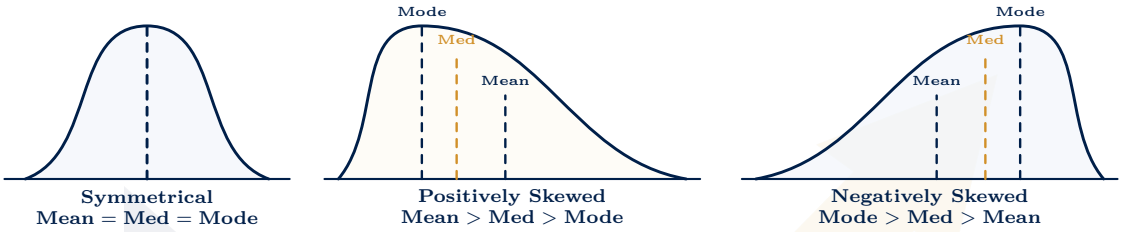
Answer: The missing frequencies are **x = 8** and **y = 7**.

EMPIRICAL RELATIONSHIP | Connecting Mean, Median and Mode

SUPPORTING SKILL

Module 19 | Distribution Shapes: Symmetrical vs. Skewed

In statistical distributions, the relative positions of Mean, Median, and Mode depend on the symmetry of the frequency curve:



Symmetrical
Mean = Med = Mode

Positively Skewed
Mean > Med > Mode

Negatively Skewed
Mode > Med > Mean

Module 20 | Karl Pearson's Empirical Formula

For a moderately skewed distribution, an often-used **approximate empirical relationship** connects the three measures. It is not an exact identity:

The Golden Empirical Equation:

$$\text{Mode} \approx \mathbf{3}\,\text{Median} - \mathbf{2}\,\text{Mean}$$

Alternative Equivalent Forms:

- $\mathbf{3}\,\text{Median} \approx \text{Mode} + \mathbf{2}\,\text{Mean}$
- $\text{Mean} \approx \frac{\mathbf{3}\,\text{Median} - \text{Mode}}{\mathbf{2}}$
- $\text{Mean} - \text{Mode} \approx \mathbf{3}(\text{Mean} - \text{Median})$

How to use the empirical relationship:

- Treat the result as an estimate, not an exact identity.
- Use it when the problem or the course method specifically calls for this relationship.
- Keep Mean, Median, and Mode in the same units.

Module 21 | Rapid Board Exam Applications

- **Example 1:** If Mean = 20 and Median = 22, find the Mode:

$$\text{Mode} \approx 3(22) - 2(20) = 66 - 40 = \mathbf{26}$$

- **Example 2:** If Mode = 15 and Mean = 18, find the Median:

$$3\,\text{Median} = \text{Mode} + 2\,\text{Mean} = 15 + 2(18) = 15 + 36 = 51 \implies \text{Median} \approx \frac{51}{3} = \mathbf{17}$$

MASTER REFERENCE MATRIX | Central Tendencies at a Glance

REFERENCE MATRIX

Use this master comparative matrix for fast lookup and revision across all three central tendencies:

Measure	Computational Formula	Key Requirement	Sensitivity & Properties
Mean ($\bar{x}$) **Direct**	$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$	Class marks: $x_i = \frac{L+U}{2}$	Most widely used measure. **Highly sensitive to extreme values (outliers)**. Based on all observations.
Mean ($\bar{x}$) **Assumed Mean**	$\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i}$	Deviations: $d_i = x_i - a$	Shift of origin. Center-shifts data to eliminate large numbers.
Mean ($\bar{x}$) **Step-Dev.**	$\bar{x} = a + \left(\frac{\sum f_i u_i}{\sum f_i}\right) h$	$u_i = \frac{x_i - a}{h}$ Equal class sizes	Shift of origin AND scale. Often efficient when class widths are equal.
Mode	$l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) h$	**Continuous, equal widths** Modal class: class with maximum f_i	Represents the most frequent value or modal class. Its value depends on the frequency pattern; do not treat it as completely immune to unusual data.
Median	$l + \left(\frac{\frac{N}{2} - cf}{f}\right) h$	**Use continuous classes** First class with $cf > \frac{N}{2}$	Positional middle (50th percentile). Resistant to extreme values; appropriate for ordered numerical data, including skewed distributions.
Empirical Relation	Mode $\approx 3\,\text{Med} - 2\,\text{Mean}$	Approximate relation for moderately skewed data; not an identity	Gives an estimate of one measure from the other two when appropriate.

Comparison of Statistical Measures in Practical Scenarios:

- **When to prefer Mean:** Finding average per-capita income, test scores, daily temperature.
- **When to prefer Mode:** Ready-made garment sizing (most demanded shoe/shirt size), market surveys.
- **When to prefer Median:** Assessing wealth/wage distribution, poverty levels, response reaction times.

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COMPREHENSIVE PROBLEM | Solving Mean, Median & Mode on One Dataset

WORKED MODEL

Master Problem: The distribution below gives the daily wages of 50 workers in a factory. Calculate the **Mean**, **Mode**, and **Median**, and verify the Empirical Relationship:

Daily Wages (Rs.)	f_i	x_i	$u_i (h = 20)$	$f_i u_i$	cf	Key Designations
$100 - 120$	12	110	-2	-24	12	f_0 (pre-modal)
$120 - 140$	**14**	130	-1	-14	**26**	**f_1 (Modal & Median Class)**
$140 - 160$	8	**150 (a)**	0	0	34	f_2 (post-modal)
$160 - 180$	6	170	$+1$	$+6$	40	—
$180 - 200$	10	190	$+2$	$+20$	50	—
Total	**$N = 50$**	—	—	**$\sum f_i u_i = -12$**	—	—

Step 1 | Mean Calculation (Step-Deviation):

$$a = 150, \quad h = 20, \quad \sum f_i = 50, \quad \sum f_i u_i = -12$$

$$\bar{x} = a + \left(\frac{\sum f_i u_i}{\sum f_i}\right) \times h = 150 + \left(\frac{-12}{50}\right) \times 20 = 150 - 4.8 = \mathbf{145.20}$$

Step 2 | Mode Calculation:

$$\text{Modal Class} = 120 - 140 \implies l = 120, \quad f_1 = 14, \quad f_0 = 12, \quad f_2 = 8, \quad h = 20$$

$$\text{Mode} = 120 + \left(\frac{14 - 12}{2(14) - 12 - 8}\right) \times 20 = 120 + \left(\frac{2}{28 - 20}\right) \times 20 = 120 + \frac{2}{8} \times 20 = \mathbf{125.00}$$

Step 3 | Median Calculation:

$$\frac{N}{2} = \frac{50}{2} = 25 \implies \text{Median Class} = 120 - 140 \quad (\text{since } cf = 26 > 25)$$

$$l = 120, \quad cf = 12 \text{ (preceding row!)}, \quad f = 14, \quad h = 20$$

$$\text{Median} = 120 + \left(\frac{25 - 12}{14}\right) \times 20 = 120 + \frac{13 \times 20}{14} = 120 + 18.57 = \mathbf{138.57}$$

Step 4 | Verification via Empirical Rule:

$$\text{Empirical Mode} \approx 3\,\text{Median} - 2\,\text{Mean} = 3(138.57) - 2(145.20) = 415.71 - 290.40 = \mathbf{125.31}$$

The grouped Mode is 125.00, while the empirical estimate is about 125.31. Their closeness is illustrative only: an approximate relationship cannot independently prove that every calculation is correct.

BOARD MASTERY | Trap Busters, Verification & Final Checklist

EXAM STRATEGY

Module 22 | The Top 6 Board Examination Blunders (And How to Bust Them)

Trap 1: The Median cf Substitution Trap: Substituting cf of the median class.

TRAP ALERT

In the median formula, cf is **always** from the preceding class (the row above)!

Trap 2: The Discontinuous Class Interval Trap: Directly using l from an inclusive interval (e.g., $l = 21$ for $21 - 30$). Convert to continuous boundaries ($20.5 - 30.5 \implies l = 20.5$) before computing Mode or Median!

Trap 3: The Step-Deviation h Omission: Forgetting to multiply by h in $\bar{x} = a + \left(\frac{\sum f_i u_i}{\sum f_i}\right) \times h$. Always write the formula explicitly with h.

Trap 4: The Modal Neighbor Swapping Trap (f_0 vs. f_2): Swapping the previous (f_0) and next (f_2) frequencies. Remember the numerical ladder: f_0 (top) $\rightarrow$ f_1 (middle) $\rightarrow$ f_2 (bottom).

Trap 5: The Missing Frequency Equation Neglect: Forgetting that $\sum f_i = N$ provides your very first linear equation ($x + y = N - \sum$ known).

Trap 6: The Class Mark Midpoint Error: Subtracting limits instead of adding them when finding x_i. Remember: $x_i = \frac{L+U}{2}$, whereas $h = U - L$.

Module 23 | Arithmetic Verification Checks

The Risk Factor (Common Error)	The Maths Manthan Verification Hack
Did I check the assumed-mean arithmetic?	Verify $\bar{x} = a + \frac{\sum f_i d_i}{N}$. The deviation sum need not be zero unless a equals the actual mean.
Did I calculate Mode correctly?	With a unique modal class, check $l <$ Mode $< l + h$. If $l = 20$, $h = 10$ and you get 34, recheck the calculation.
Did I calculate Median correctly?	Check $l \leq$ Median $\leq l + h$ for the selected median class; a boundary value is possible.
Did I solve missing frequencies (x, y) correctly?	Missing frequencies x and y must be non-negative integers. Negative or fractional results usually indicate an error or inconsistent data.

Module 24 | The 5-Point Board Examination Pre-Submission Checklist

Final 60-Second Answer Sheet Audit:

- ☐ **Continuous Intervals Confirmed:** Did you verify that classes are continuous before applying Median or Mode formulas?
- ☐ **Formula and substitution shown:** Did you state the formula and show substitutions clearly? This helps the examiner follow your method; marks depend on the applicable marking scheme.
- ☐ **Interval Check:** Does Mode fall within the modal class (when unique)? Does Median lie between the median-class boundaries, allowing a boundary value?
- ☐ **Preceding cf Verified:** In Median, did you select cf from the row ABOVE the median class?
- ☐ **Proper Units Attached:** Are final statistical averages tagged with their respective units (e.g., marks, kg, Rs., hours)?